ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MECHATRONICS

PAPER 1 · PRACTICE D · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Mechatronics paper 1 · practice d.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Accept other correct answers, sensible project decisions and neat, recognisable sketches, circuits and charts.

QUESTION 1

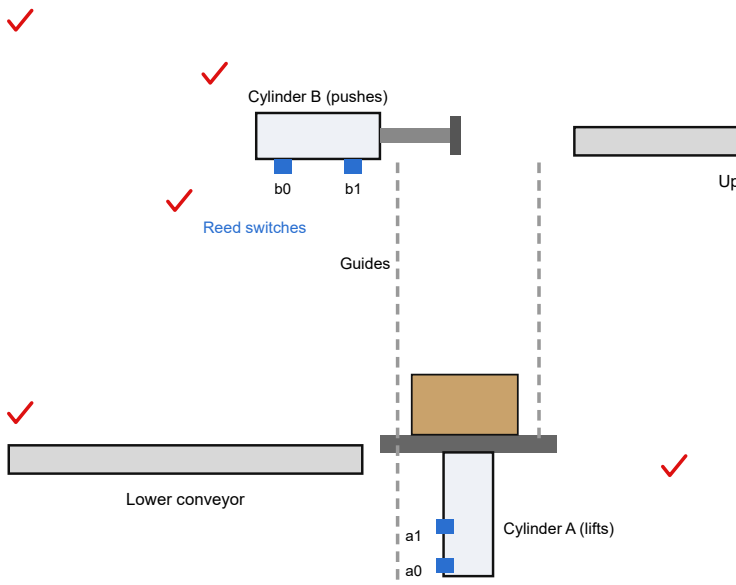
1.1 Any FOUR stages, named and explained ✓✓✓✓, e.g. identify the problem / need; research and write the design brief and specification; generate ideas and choose the best solution; detailed design and drawings; build a prototype / make the product; test and evaluate, then improve. (4)

1.2 Any SIX sensible specifications ✓✓✓✓✓✓, e.g. lift boxes up to 30 kg through 600 mm; at least 6 boxes a minute; works on 600 kPa factory air; guards and an emergency stop; fits between the existing conveyors; boxes are not damaged; quiet (silencers); cost within the budget. (6)

[10]

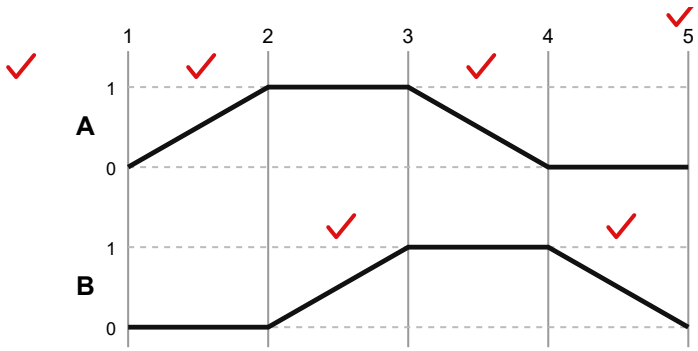
QUESTION 2

2.1



Neat freehand sketch ✓ showing: lower and upper conveyors ✓; cylinder A under the platform ✓; cylinder B at the top level pushing towards the upper conveyor ✓; reed switches on the cylinders ✓ (5)

2.2



Two diagrams (A and B) with steps 1–5 ✓; A extends from step 1 to 2 ✓; B extends from step 2 to 3 ✓; A retracts from step 3 to 4 ✓; B retracts from step 4 to 5 ✓; labels 0 / 1 and neat ✓

(6)

2.3.1 $A = \pi d^2 \div 4 = \pi \times 0{,}063^2 \div 4 = 3{,}117 \times 10^{-3}\ m^2$ ✓; $F = p \times A = 600 \times 10^3 \times 3{,}117 \times 10^{-3} = 1\ 870{,}4\ N$ ✓ (2)

2.3.2 $F_{eff} = 0{,}9 \times 1\ 870{,}4$ ✓ $= 1\ 683{,}3\ N$ ✓ (2)

2.3.3 $m_{total} = F \div g = 1\ 683{,}3 \div 9{,}81 = 171{,}6\ kg$ ✓; $m_{box} = 171{,}6 - 15 = 156{,}6\ kg$ ✓ (2)

2.4 It restricts the air flow in one direction to control (slow down) the speed of the cylinder ✓, and lets the air flow freely in the other direction through the check valve ✓. (2)

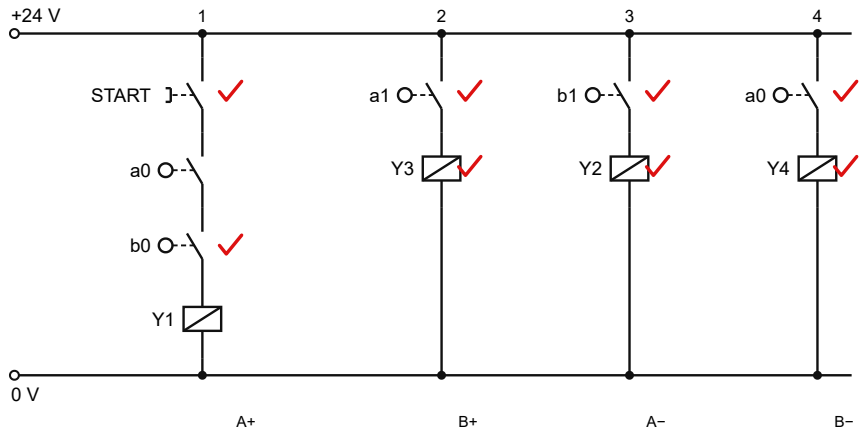
2.5 A magnet is fitted in the piston ✓; when the piston reaches the switch, the magnetic field pulls the two reed contacts together ✓, which sends a signal to the controller ✓. (3)

2.6 The cylinder is held between the air on both sides ✓, so the movement is smoother and steadier ✓ and the speed does not change as much when the load changes (no stick-slip / jerking) ✓. (3)

[25]

QUESTION 3

3.1



START and b0 (and a0) in series to Y1 (A+) ✓✓; a1 to Y3 (B+) ✓✓; b1 to Y2 (A−) ✓; a0 to Y4 (B−) ✓; correct contact and solenoid symbols ✓; labels ✓

(8)

3.2

Item	Description	Quantity	Unit cost (R)	Total (R)
1	Cylinder A (63 mm bore, 600 mm stroke)	1	3 800	3 800
2	Cylinder B (40 mm bore, 300 mm stroke)	1	1 900	1 900
3	5/2 double-solenoid valves	2	1 400	2 800
4	Reed switches	4	280	1 120
5	Air service unit (filter, regulator, soft start)	1	2 100	2 100
6	PLC and power supply	1	4 500	4 500
7	Platform, guides and frame	1	6 000	6 000

Suitable headings ✓✓; any SIX sensible items ✓✓✓✓ (2 items per mark); totals and a grand total ✓✓

(8)

3.3

Box lifter – project plan (weeks)

Task	1	2	3	4	5
Design and drawings					
Ordering parts					
Frame and platform					
Pneumatics					
Wiring and programming					
Testing and commissioning					

Week scale 1–5 along the top ✓; SIX sensible activities ✓✓ (e.g. design, ordering, frame and platform, pneumatics, wiring and programming, testing and commissioning); bars of sensible length in a logical order ✓✓✓✓; overlapping activities ✓; neat with a title ✓

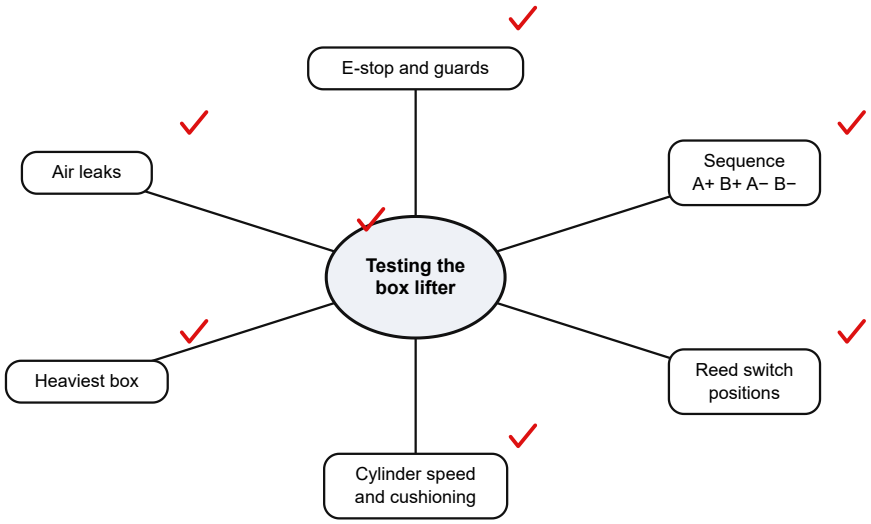
(10)

3.4 Any FOUR ✓✓✓✓: repair air leaks; use the lowest working pressure that does the job; switch the air off when the machine is not in use; use cylinders of the correct (not oversized) bore and stroke; use short tubes between the valves and cylinders; maintain the compressor. (4)

[30]

QUESTION 4

4.1



Centre topic "Testing the box lifter" ✓ with any FIVE areas ✓✓✓✓✓: emergency stop and guards; the sequence A+ B+ A− B−; reed switch positions; cylinder speeds and cushioning; lifting the heaviest box; air leaks; boxes reach the upper conveyor without damage.

(6)

4.2 Start with the flow control valves nearly closed ✓; run the cylinders and open the valves slowly ✓ until the movement is quick but smooth, without banging at the end of the stroke, then lock the settings ✓. (3)

4.3 A soft-start valve builds up the pressure slowly ✓, so cylinders do not shoot out suddenly and injure people or damage the machine ✓. (2)

4.4 Any FOUR ✓✓✓✓: tests done and results; settings (pressure, speeds, sensor positions); faults found and how they were fixed; snag list; date and signatures of the technician and client; confirmation that safety devices work. (4)

[15]

QUESTION 5

5.1

Task	How often
Drain water from the filter ✓	Daily
Check for air leaks ✓	Weekly
Check the reed switch positions and fixings ✓	Monthly
Lubricate the platform guides ✓	Monthly
Check the cylinder seals and cushioning ✓	Every 6 months

Any FOUR sensible tasks ✓✓✓✓ with a sensible frequency for each ✓✓✓✓ (8)

5.2 Make the machine safe (people clear, air and power isolated when working on it) ✓; check the 24 V at the solenoid Y3 ✓; check the wiring / connector and the solenoid coil ✓; check the air pressure at the valve and use the manual override ✓; check the tubes to cylinder B for blockage or kinks ✓; check cylinder B (seals, mechanical jam) and repair, then test again ✓. (6)

5.3 Spray soapy water on the fittings and tubes and look for bubbles ✓, or listen for hissing / use an ultrasonic leak detector while the system is under pressure ✓. (2)

5.4 Water in the air causes corrosion and washes the lubricant out of valves and cylinders ✓, which leads to sticking, wear and failure ✓. (2)

5.5 Any TWO ✓✓: worn seals are a common failure; the machine can be repaired quickly with less downtime; seal kits may take long to order; the warehouse depends on the lifter. (2)

[20]

TOTAL: 100

END OF MARKING GUIDELINE